Debugging a PHP Program (Optional)

- This is optional, further reading beyond `error_log()`.
- Two more powerful options:
  - PHPStorm: Simple and Powerful
  - Xdebug: Complex, Powerful, and works with VSCode

Use PHPStorm

- Students can request a free license from JetBrains (https://account.jetbrains.com/licenses).
    - Download JetBrains ToolBox App (https://www.jetbrains.com/toolbox-app/)
    - Download PHPStorm
    - Open the PHP Project in PHPStorm.
    - Set breakpoints and run a debugger.

Use Xdebug

Command-line Check

This command searches PHP's loaded module list for Xdebug. If it prints `xdebug`, Xdebug is installed and enabled; no output means PHP did not load it.

```
php -m | grep xdebug
```

Web-Based Check: `phpinfo()`

For more detail than `php -m`, create this temporary PHP page and run the PHP server.

- It shows PHP configuration, including the loaded `php.ini` file and installed Xdebug.
- **Delete this file once you're done: `phpinfo()` exposes detailed server configuration and should never be left reachable.**

```php
<?php
phpinfo();
?>
```

Installation of Xdebug

- Use https://xdebug.org/docs/install for the installation example.
  - For Mac, Linux, and WSL2, use `pecl`.

```
pecl install xdebug
```

- Check the installation.

```
smcho@m4 ~> php -v
PHP 8.4.11 (cli) (built: Jul 29 2025 15:30:21) (NTS)
Copyright (c) The PHP Group
Built by Homebrew
Zend Engine v4.4.11, Copyright (c) Zend Technologies
    with Xdebug v3.4.5, Copyright (c) 2002-2025, by Derick Rethans
    with Zend OPcache v8.4.11, Copyright (c), by Zend Technologies
```

Update the php.ini

- Find the location of the php.ini file.
- In this example,

```
smcho@m4 ~> php --ini
Configuration File (php.ini) Path: /opt/homebrew/etc/php/8.4
Loaded Configuration File:         /opt/homebrew/etc/php/8.4/php.ini
Scan for additional .ini files in: /opt/homebrew/etc/php/8.4/conf.d
Additional .ini files parsed:      /opt/homebrew/etc/php/8.4/conf.d/ext-opcache.ini
```

- edit "/opt/homebrew/etc/php/8.4/php.ini" to add the following.
- This is an example for Mac.
- For Windows, find the location of the DLL and adjust
- `zend_extension="C:\path\to\php\ext\php_xdebug.dll";`

```
zend_extension="xdebug.so"
xdebug.mode=debug
xdebug.start_with_request=yes
xdebug.client_host=127.0.0.1
xdebug.client_port=9003
```

VSCode

- Install "PHP Debug" extension
- Make `.vscode/launch.json`

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Listen for Xdebug",
      "type": "php",
      "request": "launch",
      "port": 9003
    }
  ]
}
```

Run XDebug

- In VSCode, in the left activity bar, choose "Run and Debug"
  - Choose "Listen for Xdebug"
- In Terminal, start the server
  - "php -S localhost:8000"
- In the Web browser, access the server
  - Open http://localhost:8000